Definitions of AI Systems Architectures and AI Training/Deployment Systems

This document introduces AI systems and definitions presented in Table 7: Five-tier AI systems complexity framework of the npj Digital Medicine article. https://doi.org/10.1038/s41746-026-02595-5

Machine learning (ML), a subfield of artificial intelligence (AI), encompasses three principal learning paradigms that differ in how a model acquires knowledge from data. In **supervised learning**, the model is trained on labelled examples, that is, input data paired with known outcomes, and learns to map inputs to outputs; this is the most widely used paradigm for clinical prediction tasks. In **unsupervised learning**, the model identifies hidden structure or patterns in unlabelled data without predefined outcomes and is commonly applied to tasks such as patient stratification and anomaly detection. In **reinforcement learning**, an agent learns by interacting with an environment and receiving feedback in the form of rewards or penalties, progressively improving its decision-making strategy; this paradigm has been explored in clinical treatment optimization and robotic surgery.

Table 1 defines the labels used in our five-tier complexity framework for classifying healthcare AI systems, organized by increasing levels of learning capability, architectural sophistication, and clinical autonomy. Each tier is defined by the system's capacity to learn and adapt, illustrated with representative healthcare examples, and mapped to the neural network architectures and distributed systems described in Tables 2 and 3. The framework spans from Deterministic AI — rule-based systems with no machine learning component — to Autonomous-Agentic AI, which encompasses goal-directed, continuously learning systems operating across distributed clinical environments.

Deep learning extends ML through the use of neural networks composed of multiple layers of artificial neurons, enabling the extraction of increasingly abstract representations from raw data. A diverse range of deep learning architectures has been developed, each suited to different data types and clinical tasks. These architectures are not tied to a single learning paradigm; rather, they may be employed under supervised, unsupervised, or reinforcement learning frameworks, depending on the availability and labelling of training data. Table 2 defines the six neural network architectures referenced in Table 7 of the npj DM article.

Distinct from the question of how a model learns is the question of how it is deployed, that is, where and in what computational arrangement training and inference take place. This is the domain of training and deployment systems, which govern the logistical, privacy, and infrastructural aspects of operationalizing AI in real-world healthcare settings. The choice of deployment system determines whether patient data must be transferred to a central server, whether privacy can be preserved through local training, and whether AI can function in resource-constrained environments such as remote clinics or wearable devices. Table 3 shows a typology of six training and deployment architectures.

In the following, we present an updated version of the five-tier AI systems complexity table, originally published in our article in npj Digital Medicine. We also provide definitions of the labels in Tables 1, 2, and 3. All references, including a citation to our article, are provided at the end of the document.

This supplementary document was prepared and updated by Ahmed Zahlan and Pek Hooi Soh for the article: Zahlan, A., Soh, P.H. & Clarysse, B. Mapping AI startup investment and innovation in healthcare using a five-tier AI systems complexity framework. npj Digit. Med. (2026). https://doi.org/10.1038/s41746-026-02595-5.

Five-tier AI systems complexity framework (updated the original table published in Zahlan, Soh, Clarysse, npj DM, 2026)

Level	AI Systems Label	Definition	Examples of Adoption in Healthcare	Training & Deployment Systems
1	**Deterministic AI** (Low Complexity)	Rule-based or symbolic systems that follow fixed "if–then" logic without learning or adapting.	▪ Appointment scheduling tools using logic trees ▪ Clinical decision support alerts for drug–drug interactions in EHRs ▪ Triage algorithms in urgent care kiosks	None (no ML)
2	**Assistive AI** (Moderate Complexity)	Basic machine learning models that support decisions on structured data and require frequent human input or supervision.	▪ Chatbots answering FAQs (e.g., Ada Health) ▪ Early clinical coding and billing automation ▪ Logistic regression models predicting readmission risk	FNN, MLP
3	**Perceptual AI** (High Complexity)	Deep learning systems capable of interpreting unstructured data like images, audio, or free-text medical records.	▪ Radiology AI tools detecting fractures, nodules, or pneumonia on X-rays (e.g., Aidoc, Zebra Medical Vision) ▪ Dermatology apps classifying skin lesions from photographs (e.g., SkinVision) ▪ Wearable devices performing real-time ECG signal analysis with personalised baseline adaptation for arrhythmia detection (e.g., Apple Watch AFib detection)	CNN, RNN
4	**Integrative AI** (Advanced Complexity)	Multi-modal systems that combine and analyze data from multiple sources (genomic, clinical, lifestyle) with minimal human oversight.	▪ Oncology platforms combining imaging, biopsy, and genomics to recommend personalized therapies (e.g., Tempus) ▪ Chronic disease management platforms integrating longitudinal EHR and wearable biosensor data for clinical decision support (e.g., Livongo, Onduo) ▪ ICU monitoring systems predicting patient deterioration (e.g., CLEW or DeepMind's Streams)	C/G/RNN TRF
5	**Autonomous-Agentic AI** (Pioneering Complexity)	Autonomous AI systems operating within tightly defined clinical tasks to broader agentic systems capable of multi-step reasoning, tool use, and adaptive decision-making across distributed clinical environments.	▪ IDx-DR was the first autonomous AI diagnostic system to receive FDA clearance across any field of medicine, with a dedicated CPT billing code (92229) subsequently established for autonomous AI diagnostic devices	All included

FNN – Feedforward neural networks
MLP – Multi-layer Perceptrons
CNN – Convolutional Neural Networks
TRF – Transformer Models
GNN – Graph Neural Networks
ML – Machine Learning
RNN – Recurrent Neural Networks

Centralized Learning (a hub)
Decentralized Learning (a network)
Federated Learning (a global cloud with local hubs)
Edge Learning (a global hub with local devices)

Table 1: Definitions and human oversight requirements of AI systems

AI Systems	Definition of Label	References	Level of Autonomy and Oversight
Deterministic AI	Rule-based systems that follow predefined logic trees without learning from data. Outputs are fully predictable given the same inputs.	Sutton et al. (2020) distinguish between knowledge-based (rule-based) clinical decision support systems, which rely on expert-encoded medical logic, and non-knowledge-based systems that leverage AI/ML and pattern recognition.	The level corresponds to the knowledge-based category—systems that follow fixed "if–then" rules without any learning component. Transparent and explainable by design, termed as symbolic AI (OECD, 2022).
Assistive AI	AI systems that augment human decision-making by providing recommendations, predictions, or data summaries, but require human judgment for final decisions.	Beam and Kohane (2018) describe early-stage ML models (e.g., logistic regression, random forests) that automate relatively structured human decision-making tasks. These models require frequent human input and supervision, placing them at the lower end of the autonomy continuum.	This level covers basic machine learning applications where algorithms depend entirely on explicit human instructions and assumptions, capturing OECD (2022)'s framing of human-in-the-loop.
Perceptual AI	AI systems specialized in interpreting complex sensory inputs (images, speech, physiological signals) to detect patterns imperceptible to humans. These systems can analyze and quantify unstructured data to yield clinically meaningful output.	The emergence of deep learning for unstructured data is anchored in the ImageNet breakthrough by Krizhevsky et al. (2012), which demonstrated the power of convolutional neural networks for image classification. Esteva et al. (2019) and Rajkomar et al. (2019) subsequently documented the rapid adoption of deep learning architectures (CNNs, RNNs) on medical imaging and electronic health record (EHR) applications. The 'Perceptual AI' tier also corresponds functionally to the 'augmentative' category of the AMA CPT AI taxonomy (2022).	Most deep learning models historically focus on non-generative predictive inference for perceptual judgment tasks, particularly in domains like medical imaging, speech, and vision, although generative architectures are increasingly being adapted to perform similar judgment functions.

Integrative AI	AI systems that synthesize multiple data types, knowledge sources, and reasoning modalities to provide holistic insights.	Topol (2019) highlighted the growing use of AI platforms that integrate data from multiple modalities—genomic, imaging, clinical, and lifestyle—to support personalized treatment decisions. These multi-modal systems represent a significant increase in architectural complexity and reduced need for human oversight relative to single-modality models	As AI systems become more autonomous and embedded in societal decision-making, the NIST AI Risk Management Framework (AI RMF 1.0) offers guidance on responsible AI use through governance, risk management, and continuous human oversight.
Autonomous-Agentic AI	AI systems that independently plan, execute, and adapt clinical decisions with minimal human intervention, operating with goal-directed autonomy across distributed environments.	Yoo et al. (2025) characterize agentic AI systems as autonomous, continuously learning architectures capable of adaptive decision-making across distributed healthcare environments, often leveraging federated or edge computing for real-time clinical support.	This level captures the frontier of AI complexity, where systems adapt autonomously to new data streams with minimal human intervention. NIST highlights the risks of distributing cognition and decision-making across human–AI networks instead of centralizing them in one system.

Table 2: AI neural network architectures

Neural Network Model	Definition
Feedforward Neural Network (FNN)	The most fundamental neural network architecture, in which information flows in one direction only — from input to output — through one or more layers of artificial neurons, with no loops or feedback connections. It serves as the conceptual building block for all deeper architectures.
Multilayer Perceptron (MLP)	A type of feedforward neural network composed of an input layer, one or more hidden layers, and an output layer, where each layer is fully connected to the next. It learns patterns through a training process called backpropagation, adjusting connection weights to minimize prediction error.
Convolutional Neural Network (CNN)	A deep learning architecture designed to process grid-like data (such as images or signals) by applying localized filters — called convolutions — to detect spatial patterns such as edges, textures, or regions of interest. CNNs are the dominant model in medical image analysis.
Graph Neural Network (GNN)	A class of neural networks designed to operate on graph-structured data, in which entities (nodes) and their relationships (edges) are explicitly modelled. GNNs learn by iteratively aggregating information from each node's neighbors, making them well-suited to molecular, genomic, and patient-network data.
Recurrent Neural Network (RNN)	A neural network architecture designed for sequential or time-series data. Unlike feedforward networks, RNNs maintain a hidden state that carries information across time steps, allowing the model to capture temporal dependencies. Variants such as Long Short-Term Memory (LSTM) networks address the challenge of learning long-range patterns.
Transformer (TRF)	A neural network architecture that processes entire input sequences simultaneously using a mechanism called self-attention, which dynamically weights the relevance of each element in the sequence to every other element. Transformers have become the dominant architecture for natural language processing and are increasingly used for clinical text, medical imaging, and longitudinal data.

Table 3: A typology of AI training and deployment architectures

Table 3 defines six training and deployment architectures, ordered from most centralised to most distributed, characterising where data resides, how model updates are transmitted, and the privacy and governance implications of each configuration in healthcare contexts.

Training/Deployment Architecture	Definition	Data Location & Movement	Privacy & Governance Profile	Relevance to Healthcare AI
Centralized Learning	All data from participating sites is pooled into a single central repository and the AI model is trained in one location. The trained model is then deployed back to clinical users.	All raw patient data transferred to and stored in one central server or cloud.	Highest data exposure risk. Raises significant concerns around data governance, consent, and cross-border transfer of sensitive patient records.	Standard approach for most commercial healthcare AI platforms (e.g., cloud-hosted diagnostic tools). Computationally efficient but requires robust institutional data-sharing agreements.
Hierarchical Learning	A multi-tier architecture in which devices train locally and submit updates to intermediate edge servers, which in turn aggregate and submit to a central cloud server. Combines elements of centralised and federated designs.	Raw data remains local; intermediate model updates travel from devices to edge nodes to central cloud.	Moderate. Raw data does not leave local sites, but aggregated updates traverse multiple tiers, each representing a potential point of interception.	Well-suited to healthcare systems spanning clinics, hospitals, and national registries. Balances privacy preservation with the need for global model improvement across care settings.
Federated Learning	The AI model is trained locally at each participating institution or device. Only model parameters — not raw data — are shared with a central aggregation server, which combines them into an updated global model.	Raw data never leaves the local institution. Only encrypted model weights or gradients are transmitted.	High privacy preservation. Meets most healthcare data governance requirements. Does not fully eliminate re-identification risk from model updates alone.	The most clinically validated distributed architecture. Applied in multi-site studies for diabetic retinopathy, COVID-19 detection, and oncology imaging. Supported by Rieke et al. (2020) as the primary framework for privacy-preserving digital health.

Training/Deployment Architecture	Definition	Data Location & Movement	Privacy & Governance Profile	Relevance to Healthcare AI
Decentralized Learning	Multiple sites train the AI model collaboratively without any central coordinating server. Each node exchanges model updates directly with neighboring peers in a peer-to-peer network; aggregation occurs at each node independently.	Raw data remains local. Model updates are shared directly between peer nodes with no central repository.	High resilience and privacy. Eliminates the single point of failure and central trust dependency present in federated learning, but introduces complexity in ensuring consistency across nodes.	Emerging in healthcare contexts where institutional trust is distributed, and no single coordinating body is acceptable (e.g., cross-national consortia). Less mature clinically than federated learning.
Edge Learning	AI inference — and in some cases, model adaptation — is performed on devices at the network edge, such as wearables, smartphones, or bedside monitors, rather than in a centralized cloud or data centre.	Data is processed and retained entirely on the edge device. No transmission to external servers during inference.	Strongest local privacy. Data never leaves the device, supporting compliance in point-of-care and remote monitoring contexts.	Enables real-time clinical decision support with minimal latency. Applied in continuous ECG monitoring, wearable biosignal analysis, and bedside deterioration alerts. Full on-device training remains largely experimental; most clinical deployments perform inference only.
Split Learning	The neural network is partitioned into two sections. Each client trains the initial layers up to a designated cut layer and transmits only the intermediate outputs — not raw data — to a server that completes the remaining layers.	Raw data stays on the client. Only intermediate layer activations (cut layer outputs) are transmitted to the server.	Moderate-to-high. Raw data is protected, but cut layer outputs may carry residual information that could theoretically be inverted. Privacy guarantees are weaker than federated learning.	Suited to resource-constrained clinical devices that cannot run a full model locally (e.g., imaging equipment, bedside monitors). Applied in health informatics for distributed deep learning. Less clinically validated than federated learning.

Note: This typology covers the six training and deployment architectures most relevant to healthcare AI research and practice. Additional architectures documented in the broader distributed ML literature — including transfer learning, blockchain-enabled federated learning, and cloud-fog-edge continuum models — introduce further trade-offs between privacy, computational cost, and data governance that are beyond the scope of the present study.

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